

UNIT 4

What is HTML (Introduction to HTML)

4.0 Learning Outcomes

After completing this module, you will be able to:

- a. Create webpages using HTML.

4.1 Introduction

This unit leads you through the basics of Hyper Text Markup Language (HTML). HTML is the building block for web pages. You will learn to use HTML to author an HTML page to display in a web browser.

4.2 What is an HTML File?

HTML is a format that tells a computer how to display a web page. The documents themselves are plain text files with special "tags" or codes that a web browser uses to interpret and display information on your computer screen.

- HTML stands for Hyper Text Markup Language
- An HTML file is a text file containing small markup tags
- The markup tags tell the Web browser how to display the page
- An HTML file must have an htm or html file extension

A Simple HTML Document

Example

```
<!DOCTYPE html>
<html>
<head>
<title>Page Title</title>
</head>
<body>

<h1>My First Heading</h1>
<p>My first paragraph.</p>

</body>
</html>
```

Example Explained

- The `<!DOCTYPE html>` declaration defines that this document is an HTML5 document
- The `<html>` element is the root element of an HTML page
- The `<head>` element contains meta information about the HTML page
- The `<title>` element specifies a title for the HTML page (which is shown in the browser's title bar or in the page's tab)
- The `<body>` element defines the document's body, and is a container for all the visible contents, such as headings, paragraphs, images, hyperlinks, tables, lists, etc.
- The `<h1>` element defines a large heading
- The `<p>` element defines a paragraph

4.3 What is an HTML Element?

An HTML element is defined by a start tag, some content, and an end tag:
`<tagname>Content goes here...</tagname>`

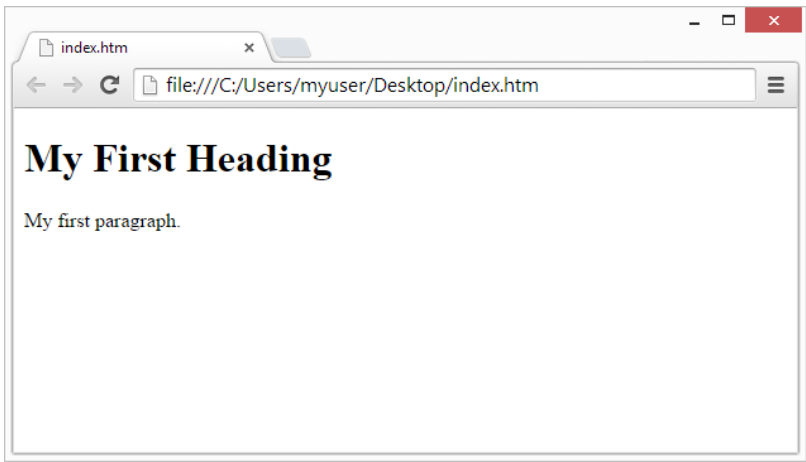
The HTML **element** is everything from the start tag to the end tag:
`<h1>My First Heading</h1>`
`<p>My first paragraph.</p>`

Start tag	Element content	End tag
<code><h1></code>	My First Heading	<code></h1></code>
<code><p></code>	My first paragraph.	<code></p></code>
<code>
</code>	<i>none</i>	<i>none</i>

Note: Some HTML elements have no content (like the `<br>` element). These elements are called empty elements. Empty elements do not have an end tag!

Web Browsers

- The purpose of a web browser (Chrome, Edge, Firefox, Safari) is to read HTML documents and display them correctly.
- A browser does not display the HTML tags, but uses them to determine how to display the document:



HTML Page Structure

Below is a visualization of an HTML page structure:

```
<html>
<head>
<title>Page title</title>
</head>
<body>
<h1>This is a heading</h1>
<p>This is a paragraph.</p>
<p>This is another paragraph.</p>
</body>
</html>
```

Note: Only the content inside the <body> section will be displayed in a browser.

4.4 Creating Webpage using HTML

Web pages can be created and modified by using professional HTML editors.

However, for learning HTML it is recommended a simple text editor like Notepad (PC) or TextEdit (Mac). Using a simple text editor is a good way to learn HTML. Follow the steps below to create your first web page with Notepad or TextEdit.

Step 1: Open Notepad (PC)

Windows 8 or later:

Open the **Start Screen** (the window symbol at the bottom left on your screen). Type **Notepad**.

Windows 7 or earlier:

Open **Start > Programs > Accessories > Notepad**

Step 1: Open TextEdit (Mac)

Open **Finder > Applications > TextEdit**

Also change some preferences to get the application to save files correctly. In **Preferences > Format >** choose "**Plain Text**"

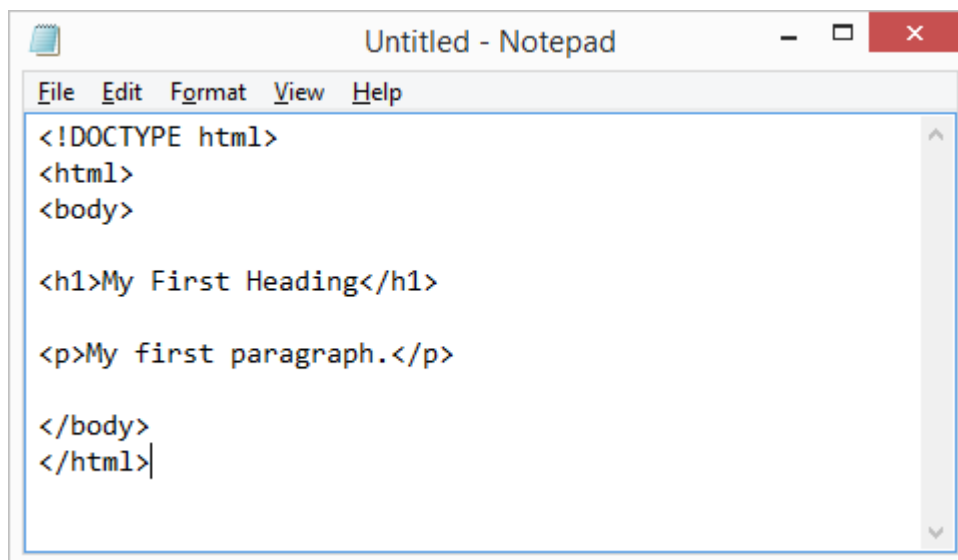
Then under "Open and Save", check the box that says "Display HTML files as HTML code instead of formatted text".

Then open a new document to place the code.

Step 2: Write Some HTML

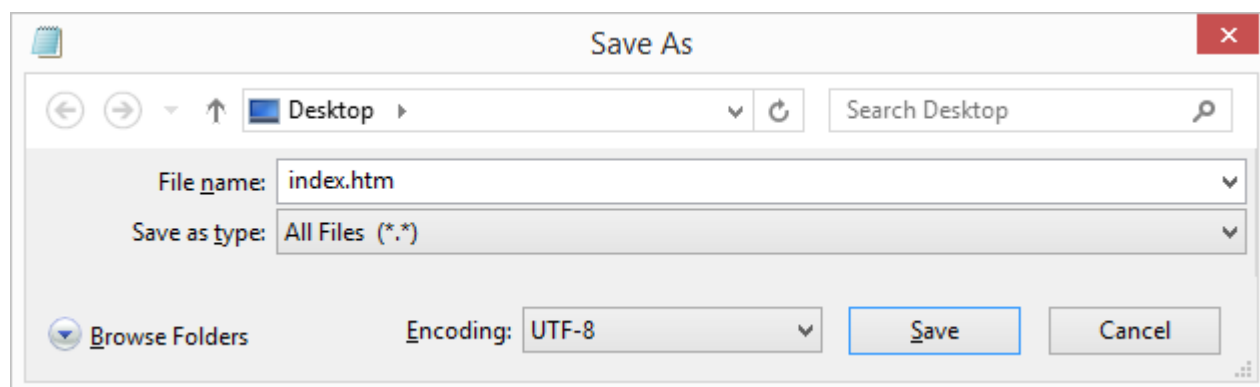
Write or copy the following HTML code into Notepad:

```
<!DOCTYPE html>
<html>
<body>
<h1>My First Heading</h1>
<p>My first paragraph.</p>
</body>
</html>
```

**Step 3: Save the HTML Page**

Save the file on your computer. Select **File > Save as** in the Notepad menu.

Name the file "**index.htm**" and set the encoding to **UTF-8** (which is the preferred encoding for HTML files).

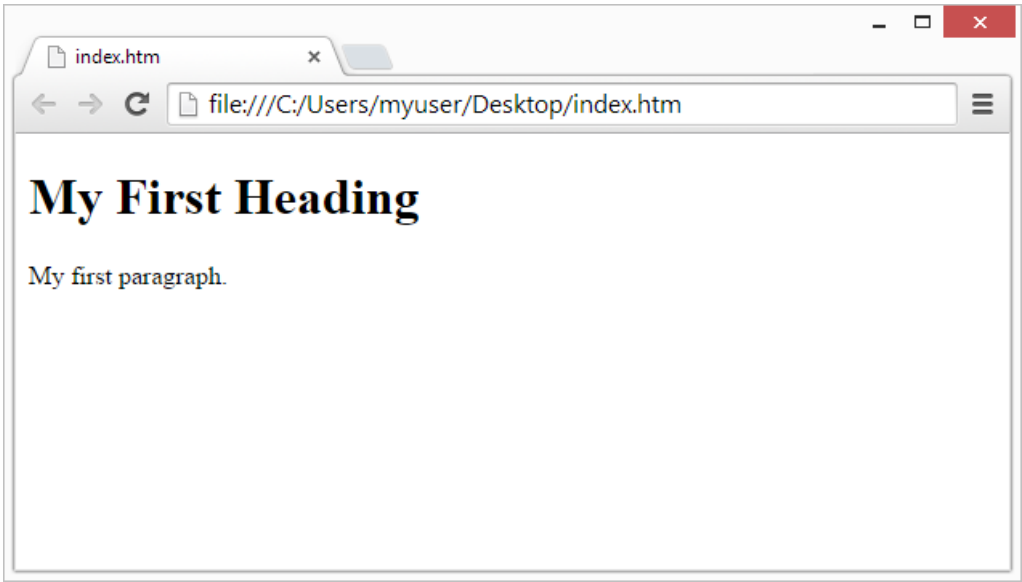


Tip: You can use either .htm or .html as file extension. There is no difference, it is up to you.

Step 4: View the HTML Page in Your Browser

Open the saved HTML file in your favorite browser (double click on the file, or right-click - and choose "Open with").

The result will look much like this:



4.5 Basic HTML Tags

The most important tags in HTML are tags that define headings, paragraphs and line breaks.

Basic HTML Tags Tag	Description
<html>	Defines an HTML document
<body>	Defines the document's body
<h1> to <h6>	Defines header 1 to header 6
<p>	Defines a paragraph
 	Inserts a single line break
<hr>	Defines a horizontal rule
<!-->	Defines a comment

4.5.1 Headings

Headings are defined with the <h1> to <h6> tags. <h1> defines the largest heading while <h6> defines the smallest.

<h1>This is a heading</h1>

<h2>This is a heading</h2>

<h3>This is a heading</h3>

<h4>This is a heading</h4>

<h5>This is a heading</h5>

<h6> This is a heading</h6>

HTML automatically adds an extra blank line before and after a heading. A useful heading attribute is align.

```
<h5 align="left">I can align headings </h5>
      <h5 align="center">This is a centered heading </h5>
<h5 align="right">This is a heading aligned to the right </h5>
```

4.5.2 Paragraphs

Paragraphs are defined with the <p> tag. Think of a paragraph as a block of text. You can use the align attribute with a paragraph tag as well.

```
<p align="left">This is a paragraph</p>
      <p align="center">this is another paragraph</p>
```

Important: You must indicate paragraphs with <p> elements. A browser ignores any indentations or blank lines in the source text. Without <p> elements, the document becomes one large paragraph. HTML automatically adds an extra blank line before and after a paragraph.

4.5.2 Line Breaks

The
 tag is used when you want to start a new line, but don't want to start a new paragraph. The
 tag forces a line break wherever you place it. It is similar to single spacing in a document.

This Code	Would Display
This is a para graph with line breaks	This is a para graph with line breaks

Note: The
 tag has no closing tag.

4.5.3 Horizontal Rule

The <hr> element is used for horizontal rules that act as dividers between sections, like this:

The horizontal rule does not have a closing tag. It takes attributes such as align and width. For instance: This Code	Would Display
<hr width="50%" align="center">	

4.5.4 Nested Tags

You may have noticed in the example above, the <body> tag also contains other tags, like the tab. When you enclose an element in with multiple tags, the last tag opened should be the first tag closed. For example:

```
<p><b><em>This is NOT the proper way to close nested tags.</p></em></b>
<p><b><em>This is the proper way to close nested tags. </em></b></p>
```

Note: It doesn't matter which tag is first, but they must be closed in the proper order.

4.6 HTML Attributes

- All HTML elements can have **attributes**
- Attributes provide **additional information** about elements
- Attributes are always specified in **the start tag**
- Attributes usually come in name/value pairs like: **name="value"**

4.6.1 The href Attribute

The `<a>` tag defines a hyperlink. The `href` attribute specifies the URL of the page the link goes to:

Example

```
<a href="https://www.facebook.com"> Facebook </a>
```

The src Attribute

The `<img>` tag is used to embed an image in an HTML page. The `src` attribute specifies the path to the image to be displayed:

Example

```

```

There are two ways to specify the URL in the `src` attribute:

1. Absolute URL - Links to an external image that is hosted on another website. Example: `src="https://www.facebook.com/images/img_logo.jpg"`.

Notes: External images might be under copyright. If you do not get permission to use it, you may be in violation of copyright laws. In addition, you cannot control external images; it can suddenly be removed or changed.

2. Relative URL - Links to an image that is hosted within the website. Here, the URL does not include the domain name. If the URL begins without a slash, it will be relative to the current page. Example: `src="img_logo.jpg"`. If the URL begins with a slash, it will be relative to the domain. Example: `src="/images/img_logo.jpg"`.

Tip: It is almost always best to use relative URLs. They will not break if you change domain.

4.6.2 The width and height Attributes

The `<img>` tag should also contain the `width` and `height` attributes, which specifies the width and height of the image (in pixels):

Example

```

```

4.6.3 The alt Attribute

The required `alt` attribute for the `<img>` tag specifies an alternate text for an image, if the image for some reason cannot be displayed. This can be due to slow connection, or an error in the `src` attribute, or if the user uses a screen reader.

Example

```

```

Example

See what happens if we try to display an image that does not exist:

```

```

4.6.4 The lang Attribute

You should always include the `lang` attribute inside the `<html>` tag, to declare the language of the Web page. This is meant to assist search engines and browsers.

The following example specifies English as the language:

```
<!DOCTYPE html>
<html lang="en">
<body>
...
</body>
</html>
```

Country codes can also be added to the language code in the `lang` attribute. So, the first two characters define the language of the HTML page, and the last two characters define the country.

The following example specifies English as the language and United States as the country:

```
<!DOCTYPE html>
<html lang="en-US">
<body>
...
</body>
</html>
```

4.7 HTML Styles

The HTML `style` attribute is used to add styles to an element, such as color, font, size, and more.

- Use the `style` attribute for styling HTML elements
- Use `background-color` for background color
- Use `color` for text colors
- Use `font-family` for text fonts
- Use `font-size` for text sizes
- Use `text-align` for text alignment

Example

I am Red

I am Blue

I am Big

The HTML Style Attribute

Setting the style of an HTML element, can be done with the `style` attribute.

The HTML `style` attribute has the following syntax:

```
<tagname style="property:value;">
```

The *property* is a CSS property. The *value* is a CSS value.

4.7.1 Background Color

The CSS `background-color` property defines the background color for an HTML element.

Example

Set the background color for a page to powderblue:

```
<body style="background-color:powderblue;">
```

```
<h1>This is a heading</h1>
```

```
<p>This is a paragraph.</p>
```

```
</body>
```

Example

Set background color for two different elements:

```
<body>

<h1 style="background-color:powderblue;">This is a heading</h1>
<p style="background-color:tomato;">This is a paragraph.</p>

</body>
```

4.7.2 Text Color

The CSS `color` property defines the text color for an HTML element:

Example

```
<h1 style="color:blue;">This is a heading</h1>
<p style="color:red;">This is a paragraph.</p>
```

4.7.3 Fonts

The CSS `font-family` property defines the font to be used for an HTML element:

Example

```
<h1 style="font-family:verdana;">This is a heading</h1>
<p style="font-family:courier;">This is a paragraph.</p>
```

4.7.4 Text Size

The CSS `font-size` property defines the text size for an HTML element:

Example

```
<h1 style="font-size:300%;">This is a heading</h1>
<p style="font-size:160%;">This is a paragraph.</p>
```

4.7.5 Text Alignment

The CSS `text-align` property defines the horizontal text alignment for an HTML element:

Example

```
<h1 style="text-align:center;">Centered Heading</h1>
<p style="text-align:center;">Centered paragraph.</p>
```

4.7.6 Background Color

You can set the background color for HTML elements:



Example

```
<h1 style="background-color:DodgerBlue;">Hello World</h1>
```

4.7.7 Text Color

You can set the color of text:

Hello World

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed diam nonummy nibh euismod tincidunt ut laoreet dolore magna aliquam erat volutpat.

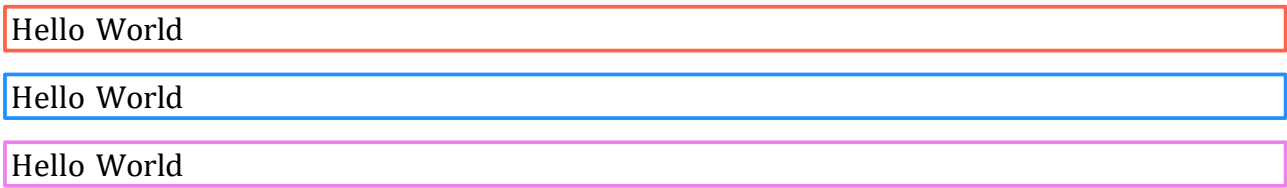
Ut wisi enim ad minim veniam, quis nostrud exerci tation ullamcorper suscipit lobortis nisl ut aliquip ex ea commodo consequat.

Example

```
<h1 style="color:Tomato;">Hello World</h1>
<p style="color:DodgerBlue;">Lorem ipsum...</p>
<p style="color:MediumSeaGreen;">Ut wisi enim...</p>
```

4.7.8 Border Color

You can set the color of borders:



Example

```
<h1 style="border:2px solid Tomato;">Hello World</h1>
<h1 style="border:2px solid DodgerBlue;">Hello World</h1>
<h1 style="border:2px solid Violet;">Hello World</h1>
```

4.8 HTML Lists

HTML provides a simple way to show unordered lists (bullet lists) or ordered lists (numbered lists).

4.8.1 Unordered Lists

An unordered list is a list of items marked with bullets (typically small black circles). An unordered list starts with the <code></code> tag. Each list item starts with the <code></code> tag. This Code	Would Display
<code> Coffee Milk </code>	☐ Coffee ☐ Milk

4.8.2 Ordered Lists

An ordered list is also a list of items. The list items are marked with numbers. An ordered list starts with the <code></code> tag. Each list item starts with the <code></code> tag. This Code	Would Display
<code> Coffee Milk </code>	1. Coffee 2. Milk

4.8.3 Definition Lists

Definition lists consist of two parts: a term and a description . To mark up a definition list, you need three HTML elements; a container <code><dl></code> , a definition term <code><dt></code> , and a definition description <code><dd></code> . This Code	Would Display
<code><dl> <dt>Cascading Style Sheets</dt> <dd>Style sheets are used to provide presentational suggestions for documents marked up in HTML. </dd> </dl></code>	Cascading Style Sheets Style sheets are used to provide presentational suggestions for documents marked up in HTML.

4.9 Tables

Tables are defined with the <table> tag. A table is divided into rows (with the <tr> tag), and each row is divided into data cells (with the <td> tag). The letters td stands for table data, which is the content of a data cell. A data cell can contain text, images, lists, paragraphs, forms, horizontal rules, tables, etc. This Code	Would Display
<table> <tr> <td>row 1, cell 1</td> <td>row 1, cell 2</td> </tr> <tr> <td>row 2, cell 1</td> <td>row 2, cell 2</td> </tr> </table>	row 1, cell 1 row 1, cell 2 row 2, cell 1 row 2, cell 2

4.9.1 Tables and the Border Attribute

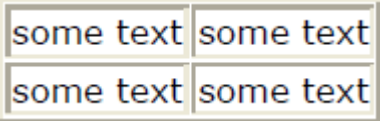
To display a table with borders, you will use the border attribute. This Code	Would Display		
<table border="1"> <tr> <td>Row 1, cell 1</td> <td>Row 1, cell 2</td> </tr> </table>	<table><tr><td>row 1, cell 1</td><td>row 1, cell 2</td></tr></table>	row 1, cell 1	row 1, cell 2
row 1, cell 1	row 1, cell 2		

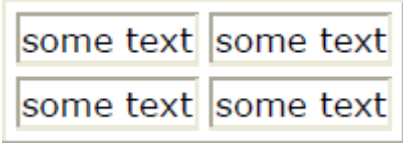
and..... This Code	Would Display		
<table border="5"> <tr> <td>Row 1, cell 1</td> <td>Row 1, cell 2</td> </tr> </table>	<table><tr><td>row 1, cell 1</td><td>row 1, cell 2</td></tr></table>	row 1, cell 1	row 1, cell 2
row 1, cell 1	row 1, cell 2		

4.9.2 Headings in a Table

Headings in a table are defined with the <th> tag. This code	Would Display						
<table border="1"> <tr> <th>Heading</th> <th>Another Heading</th> </tr> <tr> <td>row 1, cell 1</td> <td>row 1, cell 2</td> </tr> <tr> <td>row 2, cell 1</td> <td>row 2, cell 2</td> </tr> </table>	<table><tr><td>Heading</td><td>Another Heading</td></tr><tr><td>row 1, cell 1</td><td>row 1, cell 2</td></tr><tr><td>row 2, cell 1</td><td>row 2, cell 2</td></tr></table>	Heading	Another Heading	row 1, cell 1	row 1, cell 2	row 2, cell 1	row 2, cell 2
Heading	Another Heading						
row 1, cell 1	row 1, cell 2						
row 2, cell 1	row 2, cell 2						

4.9.3 Cell Padding and Spacing

The <table> tag has two attributes known as cellpadding and cellspacing. Here is a table example without these properties. These properties may be used separately or together. This Code	Would Display
<code><table border="1"> <tr> <td>some text</td> <td>some text</td> </tr> <tr> <td>some text</td> <td>some text</td> </tr> </table></code>	

Cellspacing is the pixel width between the individual data cells in the table (The thickness of the lines making the table grid). The default is zero. If the border is set at 0, the cellspacing lines will be invisible. This Code	Would Display
<code><table border="1" cellspacing="5"> <tr> <td>some text</td> <td>some text</td> </tr> <tr> <td>some text</td> <td>some text</td> </tr> </table></code>	

Cellpadding is the pixel space between the cell contents and the cell border. The default for this property is also zero. This feature is not used often, but sometimes comes in handy when you have your borders turned on and you want the contents to be away from the border a bit for easy viewing. **Cellpadding** is invisible, even with the border property turned on. **Cellpadding** can be handled in a style sheet.

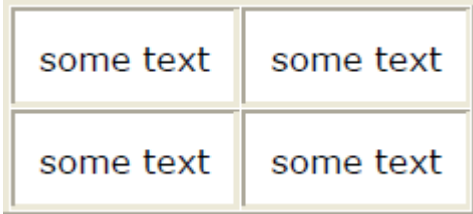
This Code	Would Display
<code><table border="1" cellpadding="10"> <tr> <td>some text</td> <td>some text</td> </tr> <tr> <td>some text</td> <td>some text</td> </tr> </table></code>	

Table Tags Tag	Description
<table>	Defines a table
<th>	Defines a table header
<tr>	Defines a table row
<td>	Defines a table cell
<caption>	Defines a table caption
<colgroup>	Defines groups of table columns
<col>	Defines the attribute values for one or more columns in a table

4.9.4 Table Size

4.9.4.1 Table Width

The width attribute can be used to define the width of your table. It can be defined as a fixed width or a relative width. A fixed table width is one where the width of the table is specified in pixels. For example, this code, `<table width="550">`, will produce a table that is 550 pixels wide. A relative table width is specified as a percentage of the width of the visitor's viewing window. Hence this code, `<table width="80%">`, will produce a table that occupies 80 percent of the screen.

This table width is 250 pixels

This table width is 50%

There are arguments in favor of giving your tables a relative width because such table widths yield pages that work regardless of the visitor's screen resolution. For example, a table width of 100% will always span the entire width of the browser window whether the visitor has a 800x600 display or a 1024x768 display (etc). Your visitor never needs to scroll horizontally to read your page, something that is regarded by most people as being very annoying.



ASSESSMENT

- I. **Direction:** For your activity, please follow the following instructions.
- 1. Visit www.sololearn.com or download **sololearn** application in your mobile phone.
 - 2. Create your own **sololearn** account.
 - 3. Make sure that you use your real name in creating your account.
 - 4. Take the course for HTML.
 - 5. At the end of each course, you will get a certificate.
 - 6. Submit the screenshot of your certificate as proof of taking the course at acpaculaba@gmail.com
With subject: CourseYearSection_LP3Assessment
Example: IT1A_ LP3Assessment
 - 7. You need to submit before receiving the Learning Packet 4.
- II. **Direction:** Create webpages using HTML. The webpages that you need to create should address different computing issues. You need to submit the activity in a form of video/vlog.

4.10 References

https://www.w3schools.com/html/html_intro.asp

https://www.w3schools.com/html/html_elements.asp

https://www.w3schools.com/html/html_attributes.asp

https://www.w3schools.com/html/html_styles.asp

https://www.w3schools.com/html/html_headings.asp

4.11 Acknowledgement

All the figures and information presented in this module were taken from the references enumerated above.

